APPENDIX A

SUSTAINABLE DEVELOPMENT GOALS

Our project, X-Attention DTI: Cross-Attention Networks for Binding Affinity Prediction, contributes to making healthcare research more efficient, accessible, and innovative through the use of artificial intelligence.

One of the key contributions of this work is in improving the process of drug discovery. Traditionally, identifying effective drug–target interactions requires extensive laboratory experiments that are time-consuming and expensive. Our model helps predict these interactions computationally, reducing the need for repeated experimental trials. Highlight the faster and more efficient drug discovery process. This supports SDG 3 – Good Health and Well-Being, as it enables quicker development of new medicines and improves healthcare outcomes.

Another important aspect of the project is the use of advanced AI techniques such as hypergraph learning, knowledge distillation, and cross-attention mechanisms. These methods allow the model to achieve high performance while using fewer computational resources. Highlight the efficient and innovative use of AI in healthcare. This contributes to SDG 9 – Industry, Innovation and Infrastructure, as it promotes the development of scalable and resource-efficient technological solutions.

The project also plays a role in education and research. It brings together concepts from computer science, artificial intelligence, and biology, encouraging interdisciplinary learning. Students and researchers can use this work as a foundation to explore modern techniques in computational biology. Highlight the encouragement of research and skill development. This supports SDG 4 – Quality Education, as it enhances knowledge and innovation in emerging scientific fields.

Overall, the project demonstrates how intelligent computational approaches can make healthcare research not only more accurate but also more efficient, innovative, and accessible for the future.